

Language-Aware Probabilistic Tokenization for Multilingual Language Models

Qixuan Shen

CCIR

Tokenization

- Tokenization as a part of model design affects
 - basic representation units
 - sequence length
 - computational efficiency-Mainly studied
 - optimization behavior
 - downstream performance (Dagan et al., 2024)

Tokenization

- Some previous studies on tokenization techniques that improves downstream performance
 - POS tagging for Chinese (Zhang and Clark, 2008)
 - For Japanese (Neubig et al., 2011)
 - Character-based models for Chinese (Li et al., 2019)
 - Sub-character representation for Chinese (Si et al., 2023)
 - Morphologically-informed tokenization for German (Hofmann et al., 2011)
- Problem: Overly specific for languages, cannot be expanded onto Multilingual LMs

Tokenization

- Multilingual tokenization approaches:
- Byte-level encoding:
 - T5 (Raffel et al., 2020) and ByT5 (Xue et al., 2022)
 - Improves downstream performance
 - Sequence is substantially longer
- Traditional approaches are still the most widely accepted.
 - BPE, Unigram
 - Improvements on this is most likely to be implementable in multilingual LMs.

Probabilistic Tokenization & Linguistic Ambiguity

- Traditionally tokenizers have 1 output sequence.
- Keeping the ambiguity results in better downstream performance
 - Unigram Subword Regularization
 - Idea: segmentation probability distribution is learnt during training
 - BPE-Dropout
 - Idea: randomly removing segmentation to allow learning of different segments
- All of the ambiguity is kept in training phase.
- Ambiguity is a greater problem in multilingual texts and code-switching hybrid languages.

Language-awareness & Representation Fusion

- Language awareness allows tokenizers to retain language preferences (Soft language information) (Pfeiffer et al., 2020).
- A linguistic premise is that in Code-switching languages like Spanglish and Hinglish, their soft language information is derived from the original languages (Poplack, 1980; Barman et al., 2014).
- Representation fusion provides a mechanism for retaining complementary information (Baltrušaitis et al., 2019) (Fedus et al., 2022).
- Thus this could help in better understanding linguistic ambiguities.

Language-Aware Probabilistic Tokenizer

- Overall it looks like this:

The deterministic baseline follows

$$X \rightarrow T \rightarrow E(T) \rightarrow \text{Transformer},$$

whereas LAPT follows

$$X \rightarrow \{T_i, p_i\}_{i=1}^k \rightarrow H \rightarrow \text{Transformer}.$$

Architecture

- Candidate set $\mathcal{C}(X) = \{T_1, T_2, \dots, T_k\}, \quad k \leq K,$
- Probability set $\mathcal{P}(X) = \{(T_1, p_1), (T_2, p_2), \dots, (T_k, p_k)\}$
- Language aware routing goes through an N-gram model.

$$\begin{aligned}\pi^{\text{prior}}(X) &= P(L \mid X) \\ &= [P(\text{en} \mid X), P(\text{es} \mid X), P(\text{hi} \mid X)].\end{aligned}$$

Architecture

- Unigram tokenizer score: $s_{\text{tok}}(T_i \mid X) = \sum_j \log P(t_{ij})$.
- Language compatibility: $s_{\text{lang}}(T_i \mid X) = \langle \pi^{\text{prior}}(X), e_i \rangle$,
- Candidate selection: $s_i^{\text{sel}} = \alpha s_{\text{tok}}(T_i \mid X) + \beta s_{\text{lang}}(T_i \mid X)$.

Architecture

- Temperature $p_i = \frac{\exp(s_i^{\text{sel}}/\tau)}{\sum_{j=1}^k \exp(s_j^{\text{sel}}/\tau)}.$

- Remove & resoftmax: $\mathcal{C}'(X) = \{(T_i, p_i) : p_i \geq \epsilon\},$

and retained probabilities are renormalized:

$$q_i = \frac{p_i}{\sum_{j \in \mathcal{C}'(X)} p_j}.$$

Probabilistic Tokenization

- Project embeddings and character alignment:

$$\tilde{h}_{ir} = \frac{\sum_j \text{ov}(a_{0r}, a_{ij}) E(t_{ij})}{\max \left(1, \sum_j \text{ov}(a_{0r}, a_{ij}) \right)}.$$

- Fuse: $H_r = \sum_i q_i \tilde{h}_{ir},$

Experiment setup

- 15.0GB of English, Hindi and Spanish from CulturaX (Nguyen et al., 2024).
- Train/test/split = 90/6.7/3.3
- GlueCoS benchmark
- XLM-R based configuration, tokenizer arm = 0.12%

Results

Task	Metric	BPE mean $\pm$ SD	LAPT mean $\pm$ SD	Δ (points)	p
NER EN–HI	Accuracy	92.857 $\pm$ 0.080	93.356 $\pm$ 0.161	+0.499	.0029
	Macro- F_1	16.049 $\pm$ 0.007	17.381 $\pm$ 1.755	+1.332	.1657
POS EN–HI FG	Accuracy	24.082 $\pm$ 0.555	23.931 $\pm$ 0.306	−0.151	.5440
	Macro- F_1	1.885 $\pm$ 0.058	1.841 $\pm$ 0.058	−0.045	.2565
POS EN–ES	Accuracy	11.712 $\pm$ 0.055	11.633 $\pm$ 0.162	−0.079	.4144
	Macro- F_1	1.222 $\pm$ 0.063	1.158 $\pm$ 0.015	−0.064	.1185
Sentiment EN–ES	Accuracy	58.707 $\pm$ 2.716	60.086 $\pm$ 2.411	+1.379	.1254
	Macro- F_1	24.651 $\pm$ 0.725	25.015 $\pm$ 0.627	+0.364	.1287

Per-language MLM loss

Test loss	BPE	LAPT
English	2.8188	2.8096
Spanish	2.4609	2.4529
Hindi	0.6159	2.0612
Macro	1.9652	2.4412

Conclusions

- Tokenization is an important part of multilingual model design
- Deterministic tokenization discards alternative linguistic structures
- LAPT preserves multiple tokenization hypotheses through language-aware probabilistic fusion.
- The approach improves some multilingual understanding tasks while showing limitations on others
- Future work should investigate better candidate generation, fusion strategies, and broader multilingual evaluation.